

Computer Science Course — ENS Year 2 (EN)

****Instructor:**** BELACEL Madani

****Institution:**** Higher Normal School of Mostaganem

****Module:**** Computer Science

****Level:**** ENS — Year 2

****Duration:**** 1h30

****Academic year:**** 2025-2026

1) Learning outcomes

By the end of this module, ENS Year 2 students will be able to:

- Apply ****object-oriented programming**** (classes, inheritance, encapsulation) in Python.
- Design ****advanced algorithms**** (sorting, search, recursion) and estimate complexity.
- Model data with the ****relational model**** and basic SQL queries.
- Understand ****Web architecture**** (HTML, CSS, JavaScript intro, HTTP).
- Use ****Git**** and collaborative workflows (branches, merge).
- Prepare ****computer science teaching**** in middle/high school (Scratch, pedagogical Python).
- Link computer science, ****internship****, and teacher ****digital portfolio****.

2) Course outline

1. Year 1 review and skill progression
2. Object-oriented programming
3. Advanced algorithms and recursion
4. Databases and SQL
5. Fundamental Web development
6. Git and software engineering intro
7. CS didactics in schools
8. Integrated Year 2 project

3) Detailed content

Chapter 1 — Review and Year 2 issues ***(10 min)***

From isolated script to ****structured software project****. Future teacher skills: algorithms, programming, digital culture, AI ethics.

Chapter 2 — OOP ***(20 min)***

Class, object, attribute, method. Constructor `__init__`. Encapsulation, simple inheritance. Example: `class Student`, `class Teacher(Student)` — school context.

Chapter 3 — Advanced algorithms *(15 min)*

Bubble sort, merge sort (overview), binary search (sorted array). Recursion: factorial, Fibonacci. Complexity comparison.

Chapter 4 — Databases *(15 min)*

Relational model: table, primary key, foreign key. SQL: `SELECT`, `WHERE`, `JOIN`, `INSERT`. Integrity and normalization (1NF intro).

Chapter 5 — Web *(15 min)*

HTML structure, CSS styling, simple JavaScript events. Client-server, forms, REST API (overview). Web accessibility.

Chapter 6 — Git and quality *(10 min)*

`git init`, `add`, `commit`, `branch`, `merge`. README, `.gitignore`. Unit tests intro (`assert`).

Chapter 7 — CS didactics *(10 min)*

Official curricula, project-based approach, Scratch → Python. Differentiation, competency assessment. Classroom AI usage charter.

Chapter 8 — Year 2 project *(5 min)*

****Deliverable:**** lightweight Web app or OOP Python module + SQLite database + Git repo + 45-minute middle-school lesson plan.

4) Assessment

Component	Weight
OOP + SQL lab	35 %
Integrated project (code + Git)	40 %
Pedagogical lesson plan	25 %

5) Bibliography

- Gutttag — **Introduction to Computation and Programming Using Python**.
- Connolly & Begg — **Database Systems** (introductory chapters).
- MDN Web Docs — HTML, CSS, JavaScript.
- Official CS teaching guidelines (national curricula).